

Test Plan

Class Being Tested: Member

Test ID	Method Being Tested	Scenario	Input(s)	Expected Output / Behaviour
TC1	add_member	Normal case, valid new member	name="Alice Wong", age=22, type="Premium", contact="alice@mail.com"	Member created with ID "M001", added to list
TC2	add_member	Duplicate contact	Add a second member with contact="alice@mail.com"	Raises ValueError, member NOT added
TC3	get_valid_int (age)	Invalid input, letters instead of number	"abc"	Program prints error, re-prompts, does not crash
TC4	get_valid_int (age)	Edge case, age exactly at minimum	1	Accepted, no error
TC5	search_member	Search by partial name, case-insensitive	"ali"	Returns list containing Alice Wong
TC6	search_member	Search with no matches	"zzz"	Returns empty list, "No matching members found" shown

Class Being Tested: FitnessClass

Test ID	Method Being Tested	Scenario	Input(s)	Expected Output / Behaviour
TC1	add_class	Normal case, valid new class	name="Yoga", instructor="Coach Lee", schedule="Mon 6PM", capacity=1	Class created with ID C001
TC2	add_class	Invalid case, zero capacity	capacity=0	Raises ValueError, class NOT added
TC3	is_full	Edge case, class exactly at capacity	capacity=1, 1 active registration	Returns True
TC4	is_full	Normal case, class below capacity	capacity=5, 2 active registrations	Returns False
TC5	remove_registration	Edge case, cancelling frees a spot	is_full() True, then cancel one registration	is_full() becomes False

Class Being Tested: Registration

Test ID	Method Being Tested	Scenario	Input(s)	Expected Output / Behaviour
TC1	register_for_class	Normal case, valid registration	Valid member_id + class_id with space available	Registration created with ID R001, status Active
TC2	register_for_class	Invalid case, class already full	Register 3rd member into capacity=2 class, already 2 active	Raises ValueError "Class is full"
TC3	register_for_class	Invalid case, duplicate active registration	Register same member for same class twice	Raises ValueError "already registered"
TC4	register_for_class	Invalid case, non-existent member ID	member_id=M999 (does not exist)	Raises ValueError "No member found"
TC5	cancel_registration	Normal case, cancel an active registration	Valid registration_id	Status becomes Cancelled, class capacity freed
TC6	cancel_registration	Invalid case, cancel an already-cancelled registration	Same registration_id cancelled twice	Raises ValueError "already cancelled"

File Handling

Test ID	Method Being Tested	Scenario	Input(s)	Expected Output / Behaviour
TC1	load_members	Edge case, first run, no file exists yet	data/members.csv does not exist	Prints info message, returns empty list, no crash
TC2	save_data / load_data	Normal case, data survives restart	Add a member, save, create a new GymSystem, load	Newly loaded list contains the same member

Menu / Input Validation

Test ID	Method Being Tested	Scenario	Input(s)	Expected Output / Behaviour
TC1	get_menu_choice	Invalid case, out-of-range menu number	"99"	Error message shown, re-prompts
TC2	get_menu_choice	Invalid case, non-numeric menu input	"abc"	Error message shown, re-prompts
TC3	get_nonempty_string	Invalid case, empty input	"" (just Enter)	Error message shown, re-prompts
TC4	Exit flow	Normal case, choosing Exit	"11"	Data auto-saved, program ends cleanly